Document 11 – Recommendation Engine Design

TUI Smart Destination Recommender

Recommendation Engine Design

Version

1.0

Purpose

This document describes the design of the recommendation engine powering the TUI Smart Destination Recommender.

The objective is to generate personalized destination recommendations that balance:

- User preferences
- Historical travel behavior
- Sustainability goals
- Destination popularity
- Congestion mitigation
- Business objectives

The recommendation engine is designed as an explainable AI system that promotes responsible tourism while maximizing traveler satisfaction.

---

1. Recommendation Objectives

The recommendation engine has five primary objectives:

Objective 1 – Personalization

Recommend destinations aligned with user interests and travel preferences.

Examples:

- Beach lovers → coastal destinations
- Nature travelers → eco destinations
- Luxury travelers → premium destinations

Objective 2 – Sustainability Optimization

Promote environmentally responsible destinations.

The engine should favor destinations with:

- Lower environmental impact
- Strong sustainability practices
- Eco-certified tourism programs

Objective 3 – Congestion Management

Reduce pressure on overcrowded destinations.

The system actively avoids directing excessive demand toward highly congested locations.

Objective 4 – Traveler Satisfaction

Maximize booking probability through relevant recommendations.

Objective 5 – Explainability

Provide transparent reasons behind every recommendation.

Example:

> Recommended because you enjoy beach vacations, the destination has excellent sustainability ratings, and current visitor levels are moderate.

---

2. Recommendation Architecture

High-Level Architecture

```
Users
 |
 ▼
User Profile Builder
 |
 ▼
Feature Engineering Layer
 |
 ▼
Recommendation Scoring Engine
 |
 ├── Preference Matching
 ├── Sustainability Scoring
 ├── Congestion Adjustment
 ├── Popularity Scoring
 |
 ▼
Ranking Engine
 |
 ▼
Top-N Recommendations
 |
 ▼
Explainability Layer
```

The architecture separates:

- Data preparation
- Scoring
- Ranking
- Explanation generation

allowing future ML models to replace individual components without redesigning the platform.

---

3. User Profile Construction

Each traveler is transformed into a structured profile.

Example:


```
{
 "user_id": 104,
 "preferred_destination_type": "Beach",
 "preferred_budget": "Medium",
 "preferred_region": "Europe",
 "travel_frequency": 4,
 "avg_trip_duration": 7
}
```


Profile information comes from:

users.csv

Provides:

- Budget preference
- Destination type
- Home country

bookings_history.csv

Provides:

- Past bookings
- Travel frequency
- Historical preferences

---

4. Feature Engineering

The recommendation engine generates recommendation features.

User Features

Examples:

preferred_budget
preferred_destination_type
booking_frequency
past_destination_count
avg_trip_duration

Destination Features

Examples:

destination_type
country
sustainability_score
congestion_score
average_cost
rating

Interaction Features

Examples:

budget_match
destination_type_match
historical_similarity

---

5. Scoring Methodology

The recommendation engine uses a weighted hybrid scoring approach.

Each destination receives multiple scores.

A. Preference Score

Measures alignment between destination attributes and user preferences.

Range:

0 – 100

Factors:

- Destination type match
- Budget match
- Region match

B. Sustainability Score

Obtained from:

sustainability_scores.csv

Example:

87 / 100

Higher values are rewarded.

C. Congestion Score

Obtained from:

congestion_scores.csv

Example:

72 / 100

Higher congestion receives penalties.

D. Popularity Score

Calculated from:

bookings_history.csv

Formula:

Popularity =
Destination Bookings /
Total Bookings

This helps balance relevance with discovery.

---

6. Hybrid Recommendation Formula

Final recommendation score:

Final Score =
(0.45 × Preference Score)
+
(0.25 × Sustainability Score)
+
(0.15 × Popularity Score)
+
(0.15 × Congestion Adjustment)

Where:

Congestion Adjustment =
100 - Congestion Score

Example:

Component	Score
Preference	90
Sustainability	85
Popularity	70
Congestion Adjustment	80

Final:

(0.45×90)
+
(0.25×85)

+
(0.15×70)
+
(0.15×80)

=
84.25

---

7. Sustainability Optimization Layer

A dedicated optimization layer ensures recommendations align with TUI sustainability goals.

Rules:

Green Boost

If:

Sustainability Score > 85

Apply:

+5% ranking boost

Eco Priority

If two destinations have similar scores:

difference < 3%

Prefer the more sustainable destination.

Sustainability Threshold

Destinations below:

50/100

receive a ranking penalty.

8. Congestion Balancing Layer

To support overtourism reduction strategies:

Low Congestion Bonus

Congestion < 40

Apply:

+5%

High Congestion Penalty

Congestion > 80

Apply:

-10%

Dynamic Demand Redistribution

If multiple destinations satisfy user preferences:

The engine prioritizes less crowded alternatives.

Example:

Instead of:

Barcelona

recommend:

Valencia

when both satisfy traveler interests.

---

9. Ranking Engine

After scoring:

```
All destinations
      |
      ▼
Calculate Final Score
      |
      ▼
Apply Sustainability Rules
      |
      ▼
Apply Congestion Rules
      |
      ▼
Sort Descending
      |
      ▼
Top 5 Recommendations
```

Output:

```
[
 {
   "destination":"Costa Rica",
   "score":91.2
 },
 {
   "destination":"Madeira",
   "score":88.7
 },
 {
   "destination":"Valencia",
   "score":86.4
 }
]
```

---

10. Explainability Layer

Explainability is a core requirement.

For every recommendation the system generates:

Recommendation Reasons

Examples:

95% match with your preferred destination type
High sustainability rating
Currently experiencing moderate visitor levels
Popular among travelers with similar interests

Example Output

```
{
 "destination":"Madeira",
 "score":88.7,
 "explanations":[
    "Matches your nature travel preferences",
    "Sustainability score of 91/100",
    "Low congestion level",
    "Popular among eco-conscious travelers"
 ]
}
```

---

11. Recommendation Flow

```
User Request
    |
    ▼
Load User Profile
    |
    ▼
Load Destination Data
    |
    ▼
Generate Features
    |
    ▼
Calculate Component Scores
    |
    ▼
```

Apply Sustainability Optimization
|
▼
Apply Congestion Balancing
|
▼
Rank Destinations
|
▼
Generate Explanations
|
▼
Return Top Recommendations

---

12. Recommendation Engine Pseudocode

```
for destination in destinations:

  preference_score = calculate_preference(
     user,
     destination
  )

  sustainability_score = destination.sustainability_score

  congestion_adjustment = (
     100 - destination.congestion_score
  )

  popularity_score = calculate_popularity(
     destination
  )

  final_score = (
     0.45 * preference_score +
     0.25 * sustainability_score +
     0.15 * popularity_score +
     0.15 * congestion_adjustment
  )

  final_score = apply_sustainability_rules(
     final_score
```

```
  )

  final_score = apply_congestion_rules(
    final_score
  )

rank_destinations()
return_top_recommendations()
```

---

13. Future ML Roadmap

The MVP uses a hybrid scoring engine.

Future versions can evolve toward AI-driven recommendations.

Phase 2

Content-Based Filtering

Using:

- Destination embeddings
- User preference vectors

Phase 3

Collaborative Filtering

Using:

- User similarity
- Booking patterns
- Behavioral clustering

Phase 4

Machine Learning Ranking

Models:

- XGBoost Ranker
- LightGBM Ranker
- CatBoost Ranking

Inputs:

- User features
- Destination features
- Historical interactions

Output:

Booking Probability

Phase 5

Generative AI Recommendations

Using LLMs to generate:

- Personalized explanations
- Travel narratives
- Sustainable travel suggestions

Conclusion

The Recommendation Engine combines personalization, sustainability optimization, congestion management, and explainable AI to deliver intelligent destination recommendations. The architecture is modular and scalable, enabling future evolution from a rule-enhanced scoring engine to a fully machine-learning-powered recommendation platform aligned with TUI's sustainability strategy.